

# Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing  
Year 1, Semester 1- 2025

## **Home Automation System**

Progress Report  
Mtr Group 02

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## 1. Introduction

The **HiveLink Smart Home Automation and Safety System** is an IoT-based smart home solution designed to enhance household safety, security, energy efficiency, and user convenience through automation. The system utilizes an ESP32 microcontroller and an Arduino Uno to integrate multiple sensors and actuators for monitoring and controlling various home functions. A DHT11 temperature and humidity sensor monitors kitchen environmental conditions and automatically activates a 12V DC ventilation fan when the temperature exceeds **30°C** or the humidity rises above **60%**. An MQ-5 gas sensor detects gas leaks and triggers a buzzer when the gas concentration reaches the threshold value of **2000**, providing an early warning to users. For improved security, an RFID-based smart door lock allows authorized access through RFID cards or tags. The system also includes an LDR sensor that automatically controls fence and rooftop lights when ambient light intensity falls below **100**, and an ultrasonic sensor that assists vehicle parking by activating a buzzer when the distance between the vehicle and the garage wall reaches **2.9 cm**. Additionally, the Blynk IoT mobile application enables real-time monitoring of temperature, humidity, and gas levels while allowing users to remotely control home lighting through a Wi-Fi connection. An LCD display provides local visualization of system status and sensor readings. By combining environmental monitoring, gas leak detection, smart ventilation, automated lighting, parking assistance, secure access control, and remote IoT management, the system offers a comprehensive and intelligent smart home solution.

## 2. System Architecture

The HiveLink Smart Home Automation and Safety System is built around two microcontrollers: the ESP32 and the Arduino Uno. The ESP32 serves as the primary controller for environmental monitoring, gas leak detection, RFID-based access control, IoT communication, and user notifications, while the Arduino Uno manages outdoor lighting automation and garage parking assistance. Together, these controllers collect sensor data, process information based on predefined threshold values, and control various output devices to improve home safety, security, energy efficiency, and user convenience.

### ESP32 MICROCONTROLLER

The ESP32 acts as the main processing and communication unit of the system. It features built-in Wi-Fi and Bluetooth capabilities, enabling seamless IoT connectivity through the Blynk mobile application. The ESP32 continuously receives data from the DHT11 temperature and humidity sensor, MQ-5 gas sensor, and RFID reader, then processes the information and controls the relay module, ventilation fan, buzzer, LCD display, and remote lighting system.

### DHT11 Temperature and Humidity Sensor (Kitchen Area)

The DHT11 sensor monitors temperature and humidity levels in the kitchen area and sends real-time data to the ESP32. The system is configured with a temperature threshold of **30°C** and a humidity threshold of **60%**. When either of these values exceeds the predefined limit, a relay module activates a **12V DC ventilation fan** to improve air circulation and maintain a safe kitchen environment. The

fan automatically turns OFF once the readings return to normal levels. Current temperature, humidity, and fan status are displayed on the LCD and Blynk application.

### **MQ-5 Gas Sensor (Kitchen Area)**

The MQ-5 gas sensor continuously monitors the kitchen for potentially hazardous gas leaks. When the gas concentration reaches or exceeds the threshold value of **2000**, the ESP32 activates a buzzer to alert occupants of a possible gas leak. At the same time, warning messages and gas sensor readings are displayed on the LCD and transmitted to the Blynk application. When gas levels remain below the threshold, the system indicates a safe operating condition.

### **LDR Module and LED Lights (Fence/Outdoor Area)**

The LDR (Light Dependent Resistor) module controls the fence or outdoor lights. At night or in low light, the LDR detects darkness and turns on the LED lights. During the day or in bright conditions, the LEDs turn off. This helps save energy and provides automatic lighting without needing manual control.

### **RFID – Based Smart Door Lock System**

The RFID access control system enhances home security by allowing only authorized users to enter the house. The RFID reader continuously scans RFID cards or tags, and when a registered card is detected, the system authenticates the user and unlocks the door. Unauthorized cards are denied access, providing a secure and convenient alternative to traditional key-based locking systems.

### **Ultrasonic Sensor and Buzzer (Garage Area)**

The ultrasonic sensor measures the distance between the car and the garage wall while parking. When the distance is less than a safe limit, the system activates a buzzer to alert the driver. This helps the driver park safely without hitting the wall.

### **LCD Display**

The LCD shows real-time sensor values, threshold levels, and system status messages, such as whether the fan is on or off, gas alert status, and general safety information.

### **Blynk IoT Application**

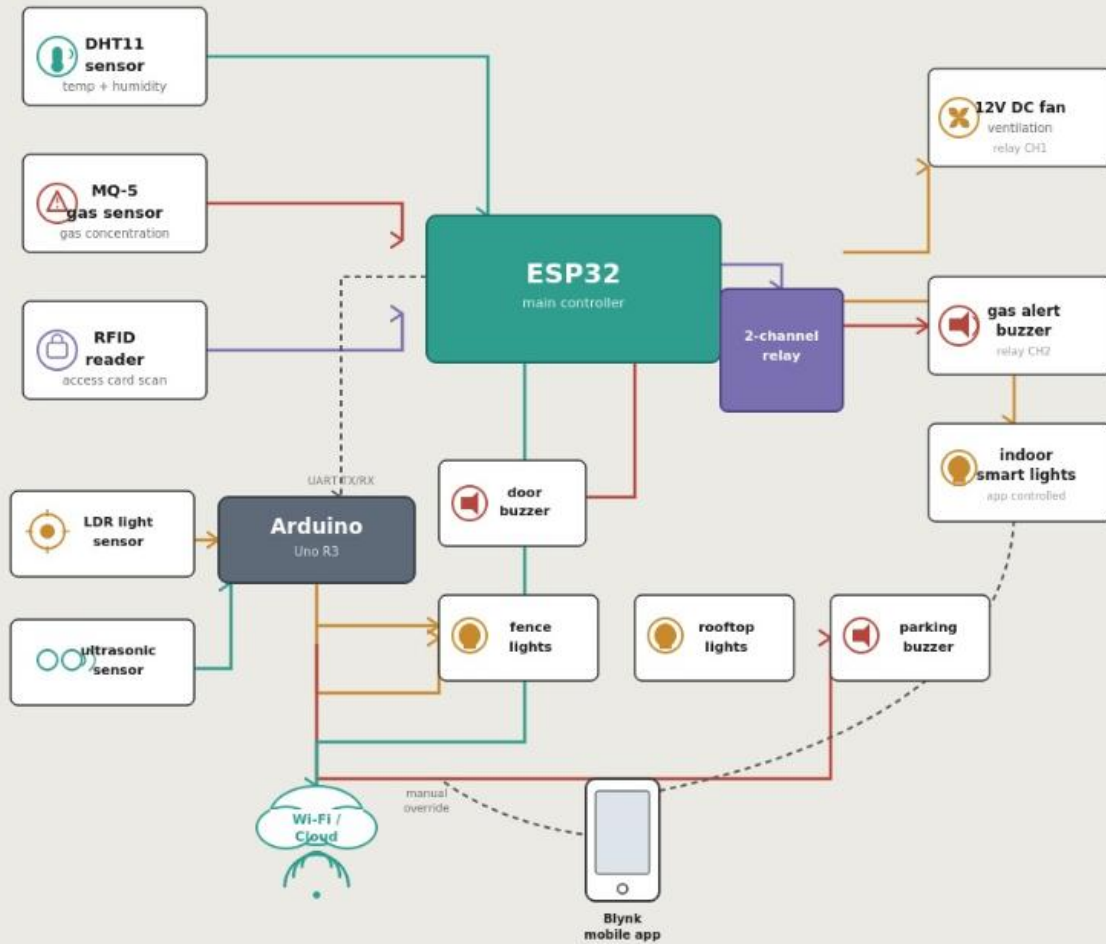
The Blynk IoT mobile application enables users to remotely control the home lighting system through the internet, allowing lights to be turned ON or OFF from any location. In addition to remote control,

the application displays real-time temperature and humidity values, gas sensor readings, and warning notifications when threshold levels are exceeded. This provides users with continuous monitoring, timely alerts, and enhanced convenience for managing home safety and automation.

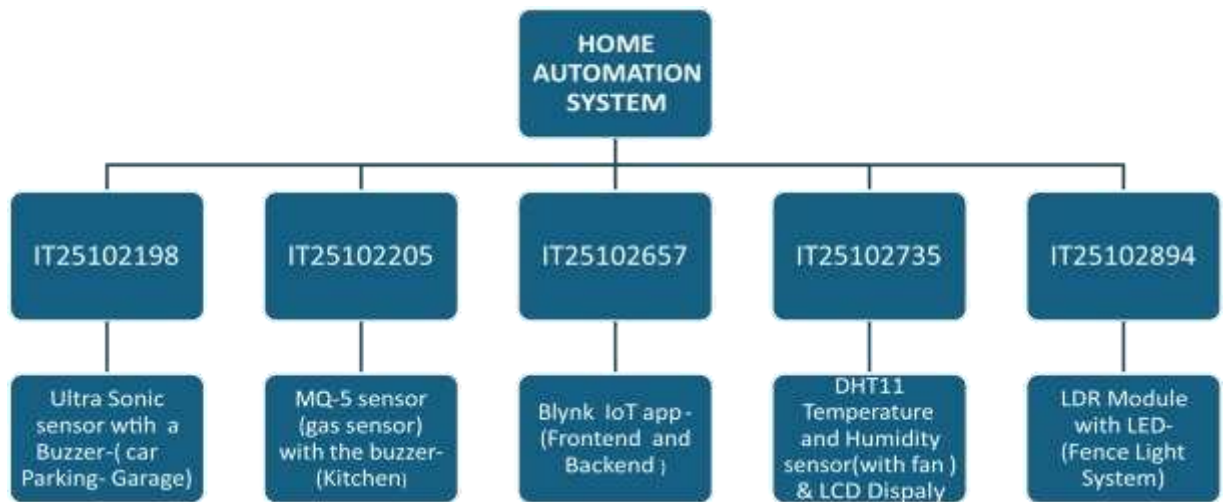
## **2. System Diagram**

### **System Diagram**

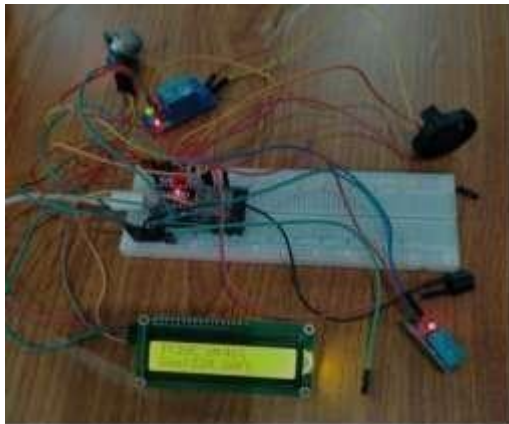
# HIVELINK SMART HOME AUTOMATION SYSTEM



### 3. Work Breakdown Structure

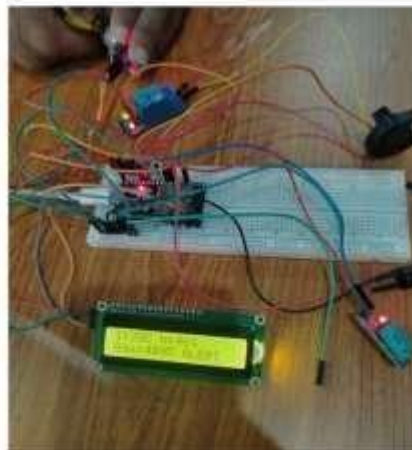


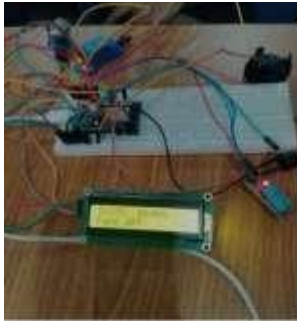
### 4. Proof of Work



MQ-5 Gas Sensor detects the gas levels

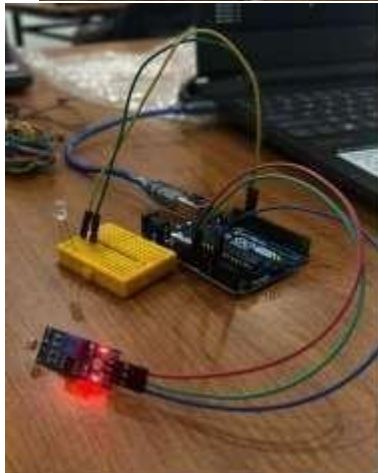
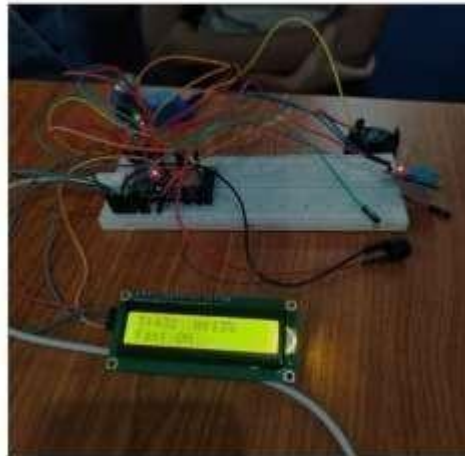
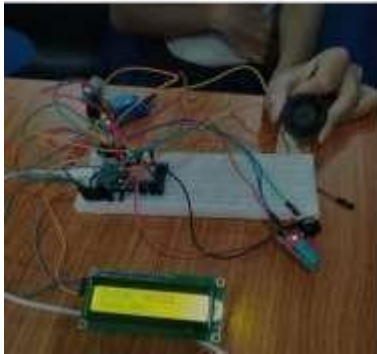
When the threshold value for gas detection ( $\text{Gas} \geq 1500$ ) is reached, automatically turns on buzzer and ALERT mode





DHT11 Humidity & Temperature sensor with Relay module ,5V DC fan and LCD

The threshold of DHT11 is  $T \geq 29$  or  $H \geq 60\%$



LDR Module with LED

Ultra Sonic Sensor with Buzzer



